

Unit: Unit 6: Systems of Linear Equations and Inequalities

Topic: Applications of Systems (Word Problems)

Name: _____

Date: _____

Advanced (Habanero)

Open Ended Questions (FRQ)

Q9. Solve the system of equations $x + 2y = 7$ and $3x - 2y = 5$. Interpret the solution in the context of the problem: Tom has 15 *to spend on ticket to a movie*. *The cost of a child's ticket is* x and the cost of an adult's ticket is y .

Q10. A farmer has 100 feet of fencing to enclose two adjacent rectangular pastures. The first pasture is x feet by y feet and the second pasture is y feet by z feet. If the total area enclosed by the two pastures is 800 square feet, write an inequality to represent this situation and solve it.

Chef's Tips

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- Tip 1: Read each question carefully before selecting an answer choice.
- Tip 2: Check to see if you can eliminate any answer choices before solving the problem.
- Tip 3: Consider drawing a graph or diagram to represent the problem.